

Apiwich Sumeksri

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Previous **Hardware Engineering Intern at Analog Devices**, where I owned a high-voltage analog signal chain from simulation through board bring-up. Electrical and Computer Engineering student at the University of Toronto, **expected graduation June 2028**, seeking a hardware engineering internship in board-level design, testing, debugging, and validation.

EDUCATION

University of Toronto

Expected Graduation: June 2028

BASc, Electrical and Computer Engineering

- 3x Dean's Honour List (Fall 2024, Winter 2025, Winter 2026)
- **Relevant coursework:** Electronic Circuits, Circuit Analysis, Computer Organization, Digital Systems, Signals & Systems, Hardware Design
- **In progress:** Analog Electronics, Digital Electronics, Computer Hardware, Operating Systems

TECHNICAL SKILLS

Languages: C, C++, Python (NumPy, Pandas, Matplotlib), Verilog, RISC-V Assembly, Java

Software & Tools: Altium Designer (schematic capture, PCB layout), LTspice, ModelSim, Quartus, STM32CubeIDE, Waveforms by Digilent

Lab & Test: Oscilloscope, multimeter, logic analyzer, production ATE, soldering and rework

Protocols: SPI, I2C, UART

EXPERIENCE

Analog Devices, Inc.

May 2026 – Aug. 2026

Hardware Engineering Intern

- Extended the output range of a precision source-measure unit (SMU) card from **$\pm 50\text{V}$ to $\pm 100\text{V}$ at 200mA** , owning the high-voltage output stage end to end. The card closes two feedback paths: voltage feedback regulating at the DUT pin, current feedback clamping at the IDAC-programmed limit.
- Built and brought up a protoboard of the extended drive stage with both feedback paths closed, demonstrating **$\pm 98\text{V}$ measured output** on $\pm 100\text{V}$ rails and correlating bench results against simulation across the full range using **oscilloscope and multimeter**.
- Designed a **cascode output stage** so no single **MOSFET** sees the full 200V VDS at 200mA , and selected the supporting high-voltage devices: complementary MOSFET pairs, 300V -class BJTs, and precision op-amps.
- Recalculated the DC biasing network for the doubled rail, re-deriving operating points across the push-pull output stage, cascode devices, and Zener/BJT bias spreader to keep every device in its linear region.
- Verified designs in LTspice, automating regression testing in **Python** (PyLTSpice) to sweep DC input across schematic revisions and flag divergence by percent-difference tolerance. Traced a DC convergence failure blocking simulation of the stage to floating nodes in the negative pull-down sub-circuit.
- Characterized the original $\pm 50\text{V}$ card on **Credence TMT ASL1000 production ATE** to establish the performance baseline for the extended design, studying existing test programs to understand the production test flow: DUT board interfacing, test sequencing, parametric measurement setup, and pass/fail limit definition.

University of Toronto Thai Student Association

Jun. 2025 – Jun. 2026

Events Associate

- Built and queried an SQL-based attendee database to group registrants by event, arrival time, and role, enabling faster check-in and smoother on-site coordination.

Flexible Radio Transceiver — Local Oscillator & Control Subsystem

Hardware Designer

- Recognized as the **only team out of 53** in the course to receive the **ECE Most Resilient Team Award** for performance in ECE295: Hardware Design and Communication.
- Designed the MCU-based local oscillator and user/computer interface daughtercard for an HF (8–16 MHz) direct-conversion radio transceiver, generating **quadrature LO signals 90 degrees apart** to drive the transmit and receive mixers and asserting the system-wide transmit-enable line.
- Owned the full **board-level design** flow: **schematic capture, PCB layout, fabrication, and hand assembly in Altium Designer**, meeting the fixed mechanical envelope, connector pinout, and stack height defined by the course-wide interface control document (ICD).
- Integrated three buses on one MCU: **I2C** to program the PLL frequency synthesizer, **SPI** to drive the OLED display, and **UART** over an FTDI USB bridge implementing the **Yaesu CAT protocol** for full PC control of frequency and TX/RX state.
- Designed the front panel around operator workflow: a rotary encoder consolidating frequency tuning and step size into a single control (turn to tune, press to change step), plus colour-coded buttons for favourite channels and operating mode.
- **Tested, debugged, and validated** the assembled board on the bench using **oscilloscope, multimeter, and logic analyzer**, verifying bus traffic and LO phase and amplitude balance before system integration with five other teams' subsystems on a shared mainboard.

Wayout 1984-Inspired FPGA Maze Game

Digital Systems Designer

- Delivered an FPGA-based first-person maze game in **Verilog** over a 3-week development cycle, integrating clock dividers, pixel-timing generators, and VGA output.
- Developed control logic using shift registers, finite state machines (FSMs), and synchronous update cycles to manage player movement, wall detection, and state transitions.
- Verified system behavior through **ModelSim** waveform analysis, register-level debugging, and cycle-accurate simulation to ensure reliable frame timing, memory access, and collision-detection accuracy.

Hexforge Arena

Digital Systems Designer

- Built a real-time top-down PvP fighting game in bare-metal **C** on the DE1-SoC with no OS or graphics libraries, directly interfacing with memory-mapped VGA for all rendering.
- Implemented a dynamic camera system tracking both players and adjusting zoom based on distance, using nearest-neighbor sampling for manual pixel scaling to the VGA buffer.
- Designed class-based combat (Warrior and Sorceress) with animation state machines, hit detection, and real-time input handling, all coordinated within a single bare-metal loop.

CPU Pipeline Performance Simulator

Systems Programmer

- Developed a cycle-accurate 5-stage CPU pipeline simulator modeling IF/ID/EX/MEM/WB execution, instruction retirement, and pipeline drain behavior.
- Implemented RAW data hazard detection and stall insertion logic to simulate pipeline bubbles and dependency-driven performance penalties.
- Analyzed pipeline performance by computing CPI, stall rate, and stage occupancy from cycle-level execution data, building an automated **Python** (NumPy, Pandas, Matplotlib) analysis flow to process results and visualize performance across workloads.